Aula 01

EXERCICIO 01 - Determine o resultado

MMC

$$\begin{array}{l|l} 3,5 & 3 \\ 1,5 & 5 \\ 1,1 & \end{array} \quad x = 15$$

1. $\left(\frac{1}{3} + \frac{-2}{5}\right) \div \frac{1}{2} \rightarrow \frac{1}{3} + \frac{-2}{5} = \frac{5-6}{15} = \frac{-1}{15} \div \frac{1}{2} = \frac{-1}{5} \cdot \frac{2}{1}$

$\frac{-2}{15}$

2. $\frac{3}{4} \times \frac{4}{3} + \frac{2}{5} \rightarrow \frac{9-16}{12} = \frac{-7}{12} \times \frac{2}{5} = \frac{24-35}{60} = \frac{-11}{60}$

$-35+24$

Primeiro divisão e multiplicação

3. $\frac{3}{7} - \boxed{\frac{2}{5} \div \frac{1}{3}} \rightarrow \frac{2}{5} \div \frac{1}{3} = \frac{2}{5} \cdot \frac{3}{1} = \frac{3}{7} - \frac{6}{5} = \frac{15-42}{35} = \frac{-27}{35}$

4. $\frac{1}{2} + \frac{1}{4} - \frac{3}{4} + \frac{1}{8} - \frac{2}{5} \rightarrow \cancel{\frac{3}{4}} - \cancel{\frac{3}{4}} + \frac{1}{8} - \frac{2}{5} = \frac{5-16}{40} = \frac{-11}{40}$

simplificar

$\frac{1}{2} + \frac{1}{4} = \frac{4+2}{8} = \frac{6}{8}^{(2)} = \frac{3}{4}$

Inverter a fração

5. $\left(\frac{1}{2}\right)^{-1} + \left(\frac{1}{3}\right)^{-1} \rightarrow \left(\frac{1}{2}\right)^{-1} = 2 + \left(\frac{1}{3}\right)^{-1} = 3 \Rightarrow 2+3=5$

$\left(\frac{2}{1}\right) = 2 \qquad \left(\frac{3}{1}\right) = 3$

6. $\frac{2}{3^{-1}} + \frac{2^{-1}}{3} \rightarrow \frac{2^{-\frac{2}{1}}}{\frac{1}{3}} + \frac{\frac{1}{2}}{\frac{3}{1}} \Rightarrow \frac{6}{1} + \frac{1}{6} = \frac{36+1}{6} = \frac{37}{6}$

$\frac{\frac{2}{1}}{\frac{1}{3}} + \frac{\frac{1}{2}}{\frac{3}{1}} = \frac{6}{1} + \frac{1}{6} =$

7. $\dfrac{4^x - 2^{2x-1}}{4^{x+1}}$ → Fatorar: $4^x - 2^{2x-1} \Rightarrow (1 - 1/2)\cdot 2^{2x}$

fatorar: $4^{x+1} = 2^{2x+2} = 2^{2x}\cdot 2^2$

$$\frac{(1-1/2)\cdot \cancel{2^{2x}}}{\cancel{2^{2x}}\cdot 2^2} = \frac{(1-1/2)}{2^2} = \boxed{\frac{1/2}{4}} = \frac{1}{4\cdot 2} = \frac{1}{8}$$

$\frac{1}{2}\cdot\frac{1}{4}$

$$\frac{2^{2x}\cdot\left(1 - 1\cdot\frac{1}{2}\right)}{2^{2x}\cdot 2^2}$$

$$\frac{4^x - \frac{2^{2x}}{2}\cdot 2^{-1}}{4^x\cdot 4^{-1}}$$

$$\frac{2^{2x} - 2^{2x}\cdot\frac{1}{2}}{2^{2x}\cdot 2^2} =$$

$$\frac{\left[1 - \frac{1}{2}\right]}{4}$$

$$\frac{\frac{1}{2}}{\frac{4}{1}} = \frac{1}{2}\cdot\frac{1}{4} = \frac{1}{8}$$

8. $2^{2^3} - (2^2)^3$ → $2^8 - 2^6 = 256 - 64 = 192$

$2^{2^3} = 2^8$

$2^{2\cdot 3 = 6} = 2^6$

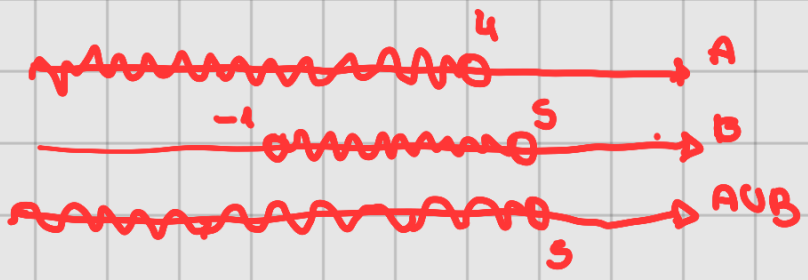
EXERCICIO 02 Escreva a união e a interseção de intervalos sob a forma de um único intervalo

a) $(3, +\infty) \cup [2, 8]$

$[2, +\infty)$

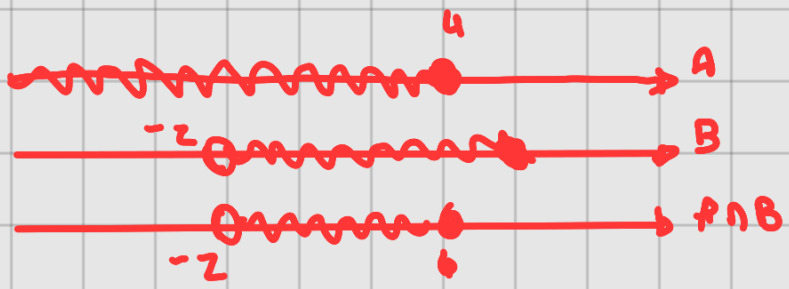
b) $(-\infty, 4) \cup (-1, 5)$

$(-\infty, 5)$



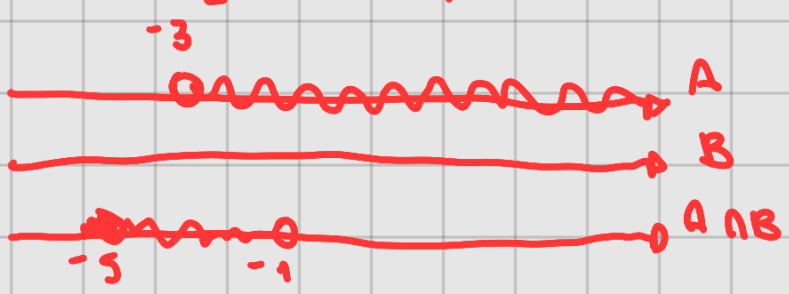
c) $(-\infty, 4] \cap (-2, 6]$

$(-2, 4]$



d) $(-3, +\infty) \cap [-5, -1)$

$(-3, -1)$



EXERCICIO 03 Determine se as afirmações são V ou F.

a) $\left(3+\frac{4}{3}\right) \geq \frac{14}{3}$ (F) $\frac{3}{1}+\frac{4}{3}=\frac{9+4}{3}=\left(\frac{13}{3}\right)$

b) $\left(\frac{1}{8} \div \frac{1}{4}\right) > \frac{1}{2}$ (F) $\frac{1}{8} \div \frac{1}{4}=\frac{1}{8}+\frac{4}{1}=\frac{4}{8}=\left(\frac{1}{2}\right)$

c) $\left(-\frac{2}{7}\right) \div \left(\frac{1}{5}-\frac{2}{3}\right) < 1$ (V) $\frac{3-10}{15}=\frac{-2}{7} \div \frac{-7}{15}=\frac{-30}{-49}=x<1$

Vai dar um número menor que 1, pois o denominador é maior que o numerador

d) $\frac{-14}{-5}=\frac{42}{15}$ (V) $\frac{-14}{-5}=\frac{14}{5}$ $\frac{42}{15}=\frac{14}{5}$ (÷3)

e) $(-\infty, 5]=\{x \in \mathbb{R} : \boxed{x \geq 5}\}$ (F) $x \geq 5 \Rightarrow [5, +\infty)$

f) $[3,7)=\{x \in \mathbb{R} : 3 < x \leq 7\}$ (F) $3<x\leq 7$ $(3, 7]$

EXERCICIO 04 - Estude o sinal da expressão.

a) $3x-1$

$3x-1>0 \rightarrow x>1/3$

$3x-1=0 \rightarrow x=1/3$

$3x-1<0 \rightarrow x<1/3$

b) $\frac{x-1}{x-2}$ $\boxed{x \neq 2}$

$\frac{x-1}{x-2} = 0 \rightarrow x-1=0 \rightarrow \boxed{x=1}$

$\frac{x-1}{x-2} > 0$

	$x-1$	$x-2$	
$x<1$	$-$	$-$	$+$
$1<x<2$	$+$	$-$	$-$
$x>2$	$+$	$+$	$+$

<0, $1<x<2$
$=0$, $x=1$
>0, $(-\infty,1)\cup(2,+\infty)$

c) $(2x+1)(x-2)$

$2x+1=0$ $\quad x=-1/2$
$x-2=0$ $\quad x=2$

	$2x+1$	$x-2$	
$x<-1/2$	$-$	$-$	$+$
$-1/2<x<2$	$+$	$-$	$-$
$x>2$	$+$	$+$	$+$

<0, $-1/2<x<2$
$=0$, $-1/2$, 2
>0, $(-\infty,-1/2)\cup(2,+\infty)$

d) $\frac{2-3x}{x+2}$ $\quad x \neq -2$ $\quad 2-3x=0$ $\quad x=2/3$

	$2-3x$	$x+2$	
$x<-2$	$+$	$-$	$-$
$-2<x<2/3$	$+$	$+$	$+$
$x>2/3$	$-$	$+$	$-$

<0, $(-\infty,-2)\cup(2/3,+\infty)$
$=0$, $2/3$
>0, $-2<x<2/3$

e) $(x-1)(x+1)(2-3x)$

$x-1=0 \rightarrow x=1$
$x+1=0 \rightarrow x=-1$
$2-3x=0 \rightarrow x=2/3$

	$x-1$	$x+1$	$2-3x$	
$x<-1$	$-$	$-$	$+$	$+$
$-1<x<2/3$	$-$	$+$	$+$	$-$
$2/3<x<1$	$+$	$+$	$+$	$+$
$x>1$	$+$	$+$	$-$	$-$

>0, $(-\infty,-1)\cup(2/3,1)$
<0, $(1,+\infty)\cup(-1,2/3)$

d) $(2x-1)(x^2+1)$:

$2x-1=0 \rightarrow x=1/2$

$x^2+1=0 \rightarrow x=\emptyset$

$<0,\ x<1/2$

$=0,\ x=1/2$

$>0,\ x>1/2$

e) $-\dfrac{x}{x+1}$ $\quad x\neq -1 \quad x=0$

inverter

	x	$x+1$	
$x<-1$	$-$	$-$	$+\ (-)$
$-1<x<0$	$-$	$+$	$-\ (+)$
$x>0$	$+$	$+$	$+\ (-)$

$>0,\ -1<x<0$

$=0,\ -1$

$<0,\ (-\infty,-1)\cup(0,+\infty)$

$x=-1,\ 1$

f) $\dfrac{1}{x+1}$ $\quad x\neq -1$

$>0,\ x<-1$

$<0,\ x>-1$

5)

a) $\dfrac{2x-1}{x+1}<0$ $\quad x\neq -1 \quad x=1/2$

	$2x-1$	$x+1$	
$x<-1$	$-$	$-$	$+$
$-1<x<1/2$	$-$	$+$	$-$
$x>1/2$	$+$	$+$	$+$

$<0,\ (-1,\ 1/2)$

b) $\frac{1-x}{3-x} \geq 0$ $\quad x \neq 3 \quad 1-x=0 \rightarrow x=1$

	$1-x$	$3-x$	
$1<x$	$-$	$-$	$+$
$1<x<3$	$-$	$+$	$-$
$x>3$	$-$	$-$	$+$

$>0, (-\infty, 1) \cup (3, +\infty)$
$=0, 1$

c) $(2x-1)(x+3)<0$

$2x-1=0 \Rightarrow x=1/2$
$x+3=0 \Rightarrow x=-3$

	$2x-1$	$x+3$	
$-3<x$	$-$	$-$	$+$
$-3<x<1/2$	$-$	$+$	$-$
$x>1/2$	$+$	$+$	$+$

$<0, (-3, 1/2)$

d) $\frac{2x-1}{x-3} > 5$ $\quad x \neq 3$

$2x-1>5$
$x=14/3$

	$2x-1$	$x-3$	
$14/3<x$	$-$	$-$	$+$
$14/3<x<3$	$+$	$-$	$-$
$x>3$	$+$	$+$	$+$

$(3, 14/3)$

e) $x(2x-1)(x+1)>0$

$x=0 \quad x=0$
$2x-1=0 \quad x=1/2$
$x+1=0 \quad x=-1$

	$2x-1$	$x+1$	x	
$-1<x$	$-$	$-$	$-$	$-$
$-1<x<0$	$-$	$+$	$-$	$+$
$0<x<1/2$	$-$	$+$	$+$	$-$
$x>1/2$	$+$	$+$	$+$	$+$

$(-1,0) \cup (1/2, +\infty)$

f) $(2x-3)(x^2+1)<0$ $\quad x^2+1=0 \quad x=\emptyset$

$2x-3=0$

$x<0 = (-\infty, 3/2) \quad x=3/2$

Exercício 5. Resolva a inequação

$1 \cdot x$

$(2x-1)\cdot(x+3)<0$

$2x^2+6x-x-3<0$

$2x^2+5x-3<0$

$S=(-3,\frac{1}{2})$

$5^2-4\cdot 2\cdot -3$

$25+24$

$\frac{-(+5)\pm\sqrt{49}}{2\cdot 2}$

$\frac{-5\pm 7}{4}$

$x'=\frac{2}{4} \quad x'=\frac{1}{2}$

$x''=\frac{-12}{4}=-3$

$$\left\{ x \in R : -3 < x < \frac{7}{2} \right\}$$